Quadratic Equation

Mind map

1. Standard form and nature of roots

- $x = \frac{-b \pm \sqrt{D}}{2a},\ D = b^2 - 4ac$
- $D > 0$ real distinct; $D = 0$ equal; $D < 0$ imaginary
- Conjugate pairs $(p \pm iq,\ p \pm \sqrt{q})$; rational roots if D is a perfect square

2. Roots, coefficients, symmetric functions

- $\alpha + \beta = -\frac{b}{a},\ \alpha\beta = \frac{c}{a},\ |\alpha - \beta| = \frac{\sqrt{D}}{|a|}$
- $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$; $\alpha^3 + \beta^3$, $\alpha^4 + \beta^4$
- Given relation between roots: combine with sum and product, eliminate

3. Formation and transformation

- $x^2 - Sx + P = 0$; cubic $x^3 - S_1x^2 + S_2x - S_3 = 0$
- Roots $f(\alpha)$: put $y = f(x)$, solve $x = g(y)$, substitute
- $a(x - \alpha)(x - \beta)$: put $x = k$; reduce powers of α by its equation

4. Newton's formula

- $S_n = p\alpha^n + q\beta^n \Rightarrow aS_{n+2} + bS_{n+1} + cS_n = 0$
- $x^2 - x - 1 = 0$: $S_{n+2} = S_{n+1} + S_n$
- $t = x + \frac{1}{x}$: $x^2 + \frac{1}{x^2} = t^2 - 2$, $x^3 + \frac{1}{x^3} = t^3 - 3t$

5. Special cases and identity

- One root 0: $c = 0, b \neq 0$; both: $b = c = 0$
- One root ∞: $a = 0, b \neq 0$; both: $a = b = 0, c \neq 0$
- Identity: $a = b = c = 0$ (more than 2 roots); $a + b + c = 0 \Rightarrow x = 1$ is a root

6. Common roots

- Both: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
- One: $(c_1a_2 - c_2a_1)^2 = (b_1c_2 - b_2c_1)(a_1b_2 - a_2b_1)$
- Subtract to eliminate α^2; imaginary roots (real coeff.) $\Rightarrow$ both common

7. Inequalities and modulus

- Wavy curve: RHS 0, factorise, sign change only at odd powers
- Double inequality: split, then intersect
- One solution set inside another: $r_1 \geq s_1,\ r_2 \leq s_2$
- $|q| < k \iff -k < q < k$; $|q| = \ldots$: split at zeroes of q

8. Polynomial and graph

- Zeroes = x-intercepts; vertex $\left(-\frac{b}{2a}, -\frac{D}{4a}\right)$
- $f > 0\ \forall x \iff a > 0,\ D < 0$; $f < 0\ \forall x \iff a < 0,\ D < 0$
- Symmetric about $x = -\frac{b}{2a}$; perfect square: $a > 0, D = 0$

9. Range of a quadratic

- $a > 0$: $\left[-\frac{D}{4a}, \infty\right)$; $a < 0$: $\left(-\infty, -\frac{D}{4a}\right]$
- Methods: complete the square, or $D \geq 0$ on $ax^2 + bx + (c - y) = 0$
- Restricted x: compare vertex (if inside) and end values

10. Range of rational functions

- $\frac{ax+b}{cx+d}$: $\mathbb{R} - \left\{\frac{a}{c}\right\}$
- $\frac{Q}{Q}, \frac{Q}{L}, \frac{L}{Q}$: cross-multiply; Case 1 $D \geq 0$; Case 2 leading coeff. $= 0$
- Common factor: cancel, find range, remove the cancelled value

11. Remainder theorem and Descartes' rule

- $f(x) \div (x - a)$: $R = f(a)$; by a quadratic: $R = ux + v$
- Positive roots $\leq$ sign changes in $f(x)$ (less by an even number)
- Negative roots: apply the rule to $f(-x)$

12. Location of roots

- Tools: D, $-\frac{b}{2a}$, sign of $af(p)$
- Both $> d$: $D \geq 0,\ af(d) > 0,\ -\frac{b}{2a} > d$; d between roots: $af(d) < 0$
- Exactly one in (d, e): $f(d)f(e) < 0$, check endpoints; IVT
- $x^4 + px^2 + q$: $t = x^2$ ($t > 0$ gives 2 values of x)

13. Higher degree equations

- $S_k = (-1)^k \frac{a_k}{a_0}$; product $= (-1)^n \frac{a_n}{a_0}$
- $\sum \alpha_i^2 = S_1^2 - 2S_2$
- $\alpha + \beta + \gamma = 0 \Rightarrow \sum \alpha^3 = 3\alpha\beta\gamma$

14. Second degree in x, y

- $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \Rightarrow$ two linear factors
- Range of y: $D_x \geq 0$; range of x: $D_y \geq 0$
- $ax^2 + 2hxy + by^2 > 0 \iff a > 0,\ h^2 < ab$

15. Substitutions and miscellaneous

- $t = x + \frac{1}{x} \Rightarrow |t| \geq 2$; $t = x^2 \Rightarrow t \geq 0$; $t = e^x \Rightarrow t > 0$
- Reciprocal equation $ax^4 + bx^3 + cx^2 \pm bx + a$: divide by x^2
- Squaring $\Rightarrow$ check roots; $p(p(x)) = 0 \iff p(x) = r$; $x = \sqrt{a + x}$; $a^3 + b^3 + c^3 - 3abc$
- Homogeneous in u, v: $t = \frac{u}{v}$. Four linear factors: pair, $t =$ common quadratic